

EXECUTIVE SUMMARY

PulsarNav AI Executive Summary

2-3 page overview of objectives, achievements, architecture, and outcomes

Project	PulsarNav AI
Repository	/Users/supryo/Desktop/pulsar
Prepared	2026-07-01
Document Class	Executive Summary
Primary Audience	Academic reviewers, aerospace mentors, flight dynamics developers

This document describes PulsarNav AI as a scientific software system: why each component exists, the mathematics behind it, how it is implemented, and how its outcomes are validated.

Executive Summary

PulsarNav AI is a research-grade software system for autonomous spacecraft navigation using millisecond pulsars as natural celestial beacons. The system demonstrates how a spacecraft can estimate its position without GPS and without continuous ground-station tracking by measuring pulsar time-of-arrival delays and solving the resulting navigation geometry.

Problem and Motivation

GPS does not provide reliable deep-space coverage because GPS satellites are Earth-orbiting transmitters with antenna patterns and signal strengths optimized for terrestrial and near-Earth users. Deep-space missions instead depend primarily on the Deep Space Network, which is accurate but scheduled, ground-dependent, and affected by communication latency. Pulsar navigation offers a complementary autonomous technique: the spacecraft observes stable astronomical sources distributed across the sky and uses timing residuals as navigation measurements.

What Was Built

- A Python scientific package for catalog parsing, coordinate conversion, timing correction, signal modeling, epoch folding, delay estimation, CRLB analysis, least-squares estimation, and Kalman filtering.
- A Next.js mission-control dashboard for interactive simulation, visualization, export, and comparison.
- A Navigation Lab API that runs LS, WLS, and EKF-style navigation studies.
- A comparative aerospace navigation engine that evaluates DSN-only, XNAV-only, and Hybrid EKF strategies under common flight dynamics.
- 3D spacecraft trajectory playback with true trajectory, estimated trajectory, error vector, pulsar directions, and mission telemetry.

Core Mathematics

The central observable is the geometric timing delay:

$$\Delta t_i = (r \cdot n_i) / c$$

where r is the spacecraft position vector, n_i is the unit direction to pulsar i , and c is the speed of light. With four or more pulsars, the system estimates three position components plus clock bias. Least squares, weighted least squares, and Kalman filtering extend this basic relation to noisy, overdetermined, and time-evolving mission cases.

Scientific Outcomes

PulsarNav AI integrates real pulsar timing file formats, models photon arrivals as a non-homogeneous Poisson process, reconstructs pulse profiles through epoch folding, estimates phase delays using CC/NLS/MLE methods, and validates navigation performance with Monte Carlo simulation. The web platform turns these algorithms into a usable research console with charts, exports, and 3D flight-state visualization.

Recommended Use

The project is best presented as a scientific software system rather than a simple code project. Its value is the full chain from astrophysical data to mathematical estimator to validated aerospace visualization.